

MSDM5004

Numerical Methods and Modeling in Science

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Lecture 5

Prof Yang Xiang

Hong Kong University of Science and Technology

Chapter 7

Iterative Methods for Solving Linear Systems

1. Iterative methods

Solve linear system $A\mathbf{x} = \mathbf{b}$

$$\sum_{j=1}^n a_{ij}x_j = b_i, \quad i = 1, 2, \dots, n.$$

by iterations

$$\mathbf{x}^{(k)} = T\mathbf{x}^{(k-1)} + \mathbf{c},$$

Advantages

- Iterative techniques are seldom used for solving linear systems of small dimension since the time required for sufficient accuracy exceeds that required for direct techniques such as Gaussian elimination.
- For large systems with a high percentage of 0 entries, however, these techniques are efficient in terms of both computer storage and computation.

sparse matrices

Recall: For a matrix of size n , the total number of operations of Gaussian elimination is $O(n^3)$.

$$\begin{bmatrix} -1 & 0 & 0 & \frac{\sqrt{2}}{2} & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{2} & 0 & -1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{2} & -1 \end{bmatrix}$$

2. Jacobi method

For each $k \geq 1$, generate the components $x_i^{(k)}$ of $\mathbf{x}^{(k)}$ from the components of $\mathbf{x}^{(k-1)}$ by

$$x_i^{(k)} = \frac{1}{a_{ii}} \left[\sum_{\substack{j=1 \\ j \neq i}}^n \left(-a_{ij} x_j^{(k-1)} \right) + b_i \right], \quad \text{for } i = 1, 2, \dots, n.$$

The i -th equation $\sum_{j=1}^n a_{ij} x_j = b_i$

Example

The linear system $A\mathbf{x} = \mathbf{b}$ given by

$$E_1 : \quad 10x_1 - x_2 + 2x_3 = 6,$$

$$E_2 : \quad -x_1 + 11x_2 - x_3 + 3x_4 = 25,$$

$$E_3 : \quad 2x_1 - x_2 + 10x_3 - x_4 = -11,$$

$$E_4 : \quad 3x_2 - x_3 + 8x_4 = 15$$

Use Jacobi's iterative technique to find approximations $\mathbf{x}^{(k)}$ to $\mathbf{x}$ starting with $\mathbf{x}^{(0)} = (0, 0, 0, 0)^t$

Solution

We first solve equation E_i for x_i , for each $i = 1, 2, 3, 4$, to obtain

the iterative scheme of the Jacobi method

$$x_1^{(k)} = \frac{1}{10}x_2^{(k-1)} - \frac{1}{5}x_3^{(k-1)} + \frac{3}{5}$$

$$x_2^{(k)} = \frac{1}{11}x_1^{(k-1)} + \frac{1}{11}x_3^{(k-1)} - \frac{3}{11}x_4^{(k-1)} + \frac{25}{11}$$

$$x_3^{(k)} = -\frac{1}{5}x_1^{(k-1)} + \frac{1}{10}x_2^{(k-1)} + \frac{1}{10}x_4^{(k-1)} - \frac{11}{10}$$

$$x_4^{(k)} = -\frac{3}{8}x_2^{(k-1)} + \frac{1}{8}x_3^{(k-1)} + \frac{15}{8}$$

$$E_1 : 10x_1 - x_2 + 2x_3 = 6,$$

$$E_2 : -x_1 + 11x_2 - x_3 + 3x_4 = 25,$$

$$E_3 : 2x_1 - x_2 + 10x_3 - x_4 = -11,$$

$$E_4 : 3x_2 - x_3 + 8x_4 = 15$$

From the initial approximation $\mathbf{x}^{(0)} = (0, 0, 0, 0)^t$ we have $\mathbf{x}^{(1)}$ given by

$$x_1^{(1)} = \frac{1}{10}x_2^{(0)} - \frac{1}{5}x_3^{(0)} + \frac{3}{5} = 0.6000,$$

$$x_2^{(1)} = \frac{1}{11}x_1^{(0)} + \frac{1}{11}x_3^{(0)} - \frac{3}{11}x_4^{(0)} + \frac{25}{11} = 2.2727,$$

$$x_3^{(1)} = -\frac{1}{5}x_1^{(0)} + \frac{1}{10}x_2^{(0)} + \frac{1}{10}x_4^{(0)} - \frac{11}{10} = -1.1000,$$

$$x_4^{(1)} = -\frac{3}{8}x_2^{(0)} + \frac{1}{8}x_3^{(0)} + \frac{15}{8} = 1.8750.$$

Additional iterates, $\mathbf{x}^{(k)} = (x_1^{(k)}, x_2^{(k)}, x_3^{(k)}, x_4^{(k)})^t$, are generated in a similar manner and are presented in Table 7.1.

Table 7.1

k	0	1	2	3	4	5
$x_1^{(k)}$	0.0000	0.6000	1.0473	0.9326	1.0152	0.9890
$x_2^{(k)}$	0.0000	2.2727	1.7159	2.053	1.9537	2.0114
$x_3^{(k)}$	0.0000	-1.1000	-0.8052	-1.0493	-0.9681	-1.0103
$x_4^{(k)}$	0.0000	1.8750	0.8852	1.1309	0.9739	1.0214

	6	7	8	9	10
	1.0032	0.9981	1.0006	0.9997	1.0001
	1.9922	2.0023	1.9987	2.0004	1.9998
	-0.9945	-1.0020	-0.9990	-1.0004	-0.9998
	0.9944	1.0036	0.9989	1.0006	0.9998

Remark: The exact solution is $x=(1,2,-1,1)^t$.

Analysis of the Jacobi method

Linear system $A\mathbf{x} = \mathbf{b}$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

write $A = D - L - U$ where

$$D = \begin{bmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & a_{nn} \end{bmatrix} \quad L = \begin{bmatrix} 0 & \cdots & \cdots & 0 \\ -a_{21} & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ -a_{n1} & \cdots & -a_{n,n-1} & 0 \end{bmatrix}$$

and

$$U = \begin{bmatrix} 0 & -a_{12} & \cdots & -a_{1n} \\ \vdots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & -a_{n-1,n} \\ 0 & \cdots & \cdots & 0 \end{bmatrix}$$

The equation $\mathbf{Ax} = \mathbf{b}$, or $(\mathbf{D} - \mathbf{L} - \mathbf{U})\mathbf{x} = \mathbf{b}$, is then transformed into

$$\mathbf{D}\mathbf{x} = (\mathbf{L} + \mathbf{U})\mathbf{x} + \mathbf{b}$$

and, if $\mathbf{D}^{-1}$ exists, that is, if $a_{ii} \neq 0$ for each i , then

$$\mathbf{x} = \mathbf{D}^{-1}(\mathbf{L} + \mathbf{U})\mathbf{x} + \mathbf{D}^{-1}\mathbf{b}$$

This results in the matrix form of the Jacobi iterative technique:

$$\mathbf{x}^{(k)} = \mathbf{D}^{-1}(\mathbf{L} + \mathbf{U})\mathbf{x}^{(k-1)} + \mathbf{D}^{-1}\mathbf{b}, \quad k = 1, 2, \dots$$

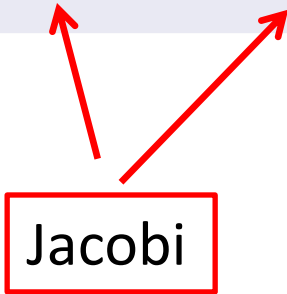
Assumption: $a_{ii} \neq 0$, for each $i = 1, 2, \dots, n$.

This formulation based on D, L, U is only for analysis.

Use the formulation on page 8 in codes.

Introducing the notation $T_j = D^{-1}(L + U)$ and $\mathbf{c}_j = D^{-1}\mathbf{b}$ gives the Jacobi technique the form

$$\mathbf{x}^{(k)} = T_j \mathbf{x}^{(k-1)} + \mathbf{c}_j$$



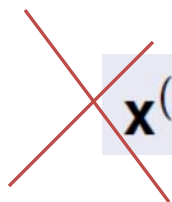
Jacobi

Reminder

Use this formula in your computation:

$$x_i^{(k)} = \frac{1}{a_{ii}} \left[\sum_{\substack{j=1 \\ j \neq i}}^n (-a_{ij} x_j^{(k-1)}) + b_i \right], \quad \text{for } i = 1, 2, \dots, n.$$

Don't use this formula in your computation:


$$\mathbf{x}^{(k)} = D^{-1}(L + U)\mathbf{x}^{(k-1)} + D^{-1}\mathbf{b},$$

3. Gauss-Seidel method

Iteration algorithm:

$$x_i^{(k)} = \frac{1}{a_{ii}} \left[- \sum_{j=1}^{i-1} (a_{ij} x_j^{(k)}) - \sum_{j=i+1}^n (a_{ij} x_j^{(k-1)}) + b_i \right]$$

Already obtained

The i-th equation $\sum_{j=1}^n a_{ij} x_j = b_i$

Example

Use the Gauss-Seidel iterative technique to find approximate solutions to

$$\begin{aligned}10x_1 - x_2 + 2x_3 &= 6, \\ -x_1 + 11x_2 - x_3 + 3x_4 &= 25, \\ 2x_1 - x_2 + 10x_3 - x_4 &= -11, \\ 3x_2 - x_3 + 8x_4 &= 15\end{aligned}$$

starting with $\mathbf{x} = (0, 0, 0, 0)^t$ and iterating until

$$\frac{\|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\|_{\infty}}{\|\mathbf{x}^{(k)}\|_{\infty}} < 10^{-3}.$$

Solution

For the Gauss-Seidel method we write the system, for each $k = 1, 2, \dots$ as

$$x_1^{(k)} = \frac{1}{10}x_2^{(k-1)} - \frac{1}{5}x_3^{(k-1)} + \frac{3}{5},$$

$$x_2^{(k)} = \frac{1}{11}x_1^{(k)} + \frac{1}{11}x_3^{(k-1)} - \frac{3}{11}x_4^{(k-1)} + \frac{25}{11},$$

$$x_3^{(k)} = -\frac{1}{5}x_1^{(k)} + \frac{1}{10}x_2^{(k)} + \frac{1}{10}x_4^{(k-1)} - \frac{11}{10},$$

$$x_4^{(k)} = -\frac{3}{8}x_2^{(k)} + \frac{1}{8}x_3^{(k)} + \frac{15}{8}.$$

When $\mathbf{x}^{(0)} = (0, 0, 0, 0)^t$, we have $\mathbf{x}^{(1)} = (0.6000, 2.3272, -0.9873, 0.8789)^t$.

Subsequent iterations give the values in Table

k	0	1	2	3	4	5
$x_1^{(k)}$	0.0000	0.6000	1.030	1.0065	1.0009	1.0001
$x_2^{(k)}$	0.0000	2.3272	2.037	2.0036	2.0003	2.0000
$x_3^{(k)}$	0.0000	-0.9873	-1.014	-1.0025	-1.0003	-1.0000
$x_4^{(k)}$	0.0000	0.8789	0.9844	0.9983	0.9999	1.0000

Because

$$\frac{\|\mathbf{x}^{(5)} - \mathbf{x}^{(4)}\|_{\infty}}{\|\mathbf{x}^{(5)}\|_{\infty}} = \frac{0.0008}{2.000} = 4 \times 10^{-4}$$

$\mathbf{x}^{(5)}$ is accepted as a reasonable approximation to the solution.

Analysis of the Gauss-Seidel method

Linear system $A\mathbf{x} = \mathbf{b}$

write $A = D - L - U$ $a_{ii} \neq 0$, for each $i = 1, 2, \dots, n$.

$$(D - L)\mathbf{x}^{(k)} = U\mathbf{x}^{(k-1)} + \mathbf{b}$$

$$\mathbf{x}^{(k)} = (D - L)^{-1}U\mathbf{x}^{(k-1)} + (D - L)^{-1}\mathbf{b},$$

Form of iteration: $\mathbf{x}^{(k)} = T_g\mathbf{x}^{(k-1)} + \mathbf{c}_g$

Gauss-Seidel

This formulation based on D, L, U is only for analysis.
Use the formulation on the previous page in codes.